

Philip B. Gottschalk

Software Engineer | Graphics, Rendering & Apple Platforms

Barnt Green, Birmingham, UK

philip@writtenbyabriton.com · +44 7737 526553

LinkedIn: <https://www.linkedin.com/in/philip157/>

GitHub: <https://github.com/tiofw/>

Portfolio: <https://tiofw.github.io/portfolio/>

Professional Profile

Experienced software engineer with 25+ years' development experience specialising in graphics, rendering, custom UI systems and Apple-platform software. Expertise in Objective-C, Metal and native macOS/iOS development, with current hands-on experience in Swift and SwiftUI.

Designed and implemented software systems from first principles, including custom rendering engines, GPU-based graphics, text-rendering technologies and complete applications, across the full lifecycle from architecture and UX through implementation, optimisation, release and maintenance.

Technical Expertise

Languages: Objective-C, Swift, C++, Metal Shading Language

Apple: iOS, macOS, SwiftUI, UIKit, AppKit, Metal, Core Graphics, Core Image, Core Text, Core Animation, CloudKit

Graphics: GPU rendering, Metal render/compute pipelines, shaders, custom rendering engines, texture management, real-time rendering

Engineering: Architecture, custom UI systems, multithreading/concurrency, memory optimisation, performance profiling, debugging

Tools: Xcode, Instruments, XCTest, Git, App Store Connect

Recent Software Development

HowLS — Swift / SwiftUI | 2026

Productivity application designed to help users manage recurring low-priority tasks without traditional deadline pressure.

- Designed and developed independently from concept through App Store release.
- Completed design, implementation, testing, visual assets and App Store materials in under two weeks.

Block Ed — Objective-C / Metal | 2025

Universal macOS and iOS application developed over approximately two years.

- Designed and implemented the application architecture, UI and rendering systems.
- Developed custom Metal shader effects and GPU render/compute pipelines.
- Built PDF export, undo/redo, CloudKit sync, autosave and backup systems.
- Adapted the interface across macOS, iPhone and iPad, localised into four languages.
- Managed the complete development and App Store release process.

Other 2026 releases: Q-Convert 2 (Swift/SwiftUI) · Colour Well (Swift/Quartz)

Professional Experience

Independent Software Engineer / Developer | 2001–Present

- Long-term independent software development encompassing applications, graphics technologies, developer tools and reusable libraries.
- Designed and implemented software from concept through release, covering architecture, UX, implementation, testing, optimisation and maintenance.
- Developed software across macOS, iOS and Windows, including custom graphics, typography, UI, document-generation and file-management technologies.

thePit.com — Chief Web Architect | c. 2000–2001

- Designed the HTML and underlying ASP architecture for the original website and developed a Java stock ticker.
- Joined during the startup phase; the company launched in October 2000 and was acquired by The Topps Company in 2001.

Sanwa International Ltd, London — Risk Management Analyst / Software Developer | 1998–2000

- Developed **Genesis**, praised as ‘a revelation’ by *Futures & OTC World* (April 1999), within the bank's risk-management department.
- Initially developed as sole programmer to replace static Excel graphs with an interactive graphical risk-management interface.
- Evolved into a database-driven system using SQL Server, with graphs generated dynamically from daily risk data.
- Rolled out to Sanwa branches in Hong Kong, New York and Tokyo.

Selected Technical Projects

Advanced Typography Library

Custom GPU-based text-rendering system supporting gradients, shadows, animated texture fills and 3D extrusion, with reusable rendering infrastructure for integration into other applications.

Video Generation Engine

Automated multilingual promotional and tutorial video generation from plain-text scripts, including screenshot capture, asset integration and video assembly.

Logo Design & Export Tool

Metal compute-based real-time logo design and asset-generation tool, automatically exporting multiple variants and application configuration files.

Other Software

Earlier work includes C++ desktop applications, games, backup software, and reusable graphics, typography, cryptography and document-generation libraries.

Education

University of York — BA Mathematics, First Class Honours | 1997